Department of Applied Data Science
DATA 298A MSDA Project I

Progress Report 2

Team 6 | Section 22 | Topic 12
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Date of report: October 22, 2026 | Evidence checked: October 21, 2026
Linear cycles covered: Cycles 3–4 (September 24 – October 21, 2026)

Neurosymbolic Verification of LLM-Generated Clinical Discharge and Treatment Plans

GitHub repository: https://github.com/Akashkumarsenthil/Clinical-Discharge-Treatment-Plan-Verification
Linear board: https://linear.app/298a-team-6-neurosymbolic/team/T6/all

Part A Project Status

Item	Planning	Investigation	Complete	Document
1 Introduction				
1.1 Project Background and Executive Summary	100%	90%	100%	100%
1.2 Project Requirements	90%	80%	90%	100%
1.3 Project Deliverables	90%	80%	90%	100%
1.4 Technology and Solution Survey	90%	80%	80%	100%
1.5 Literature Survey of Existing Research	90%	75%	70%	100%
2 Data and Project Management Plan				
2.1 Data Management Plan	95%	90%	90%	100%
2.2 Project Management Plan (2.2.1–2.2.8)	85%	70%	70%	100%
2.3 Project Resource Requirements and Plan	80%	60%	50%	100%
2.4 Project Schedule	90%	80%	80%	100%
3 Data Engineering				
3.1 Data Process	90%	85%	85%	85%
3.2 Data Collection	100%	100%	100%	85%
3.3 Data Pre-processing	100%	100%	100%	80%
3.4 Data Transformation	100%	100%	100%	80%
3.5 Data Preparation	100%	100%	100%	85%

3.6 Data Statistics	100%	100%	100%	80%
4 Model Development				
4.1 Model Proposals	85%	75%	60%	40%
4.2 Model Supports	65%	50%	30%	25%
4.3 Model Comparison and Justification	70%	60%	35%	30%
4.4 Model Evaluation Methods	80%	65%	45%	45%
4.5 First Model Implementation	40%	20%	0%	0%
4.6 Baseline Reproduction	70%	50%	10%	15%
5 Prototype and System Implementation	30%	15%	0%	0%
6 Benchmarking, Evaluation, and Failure Study	60%	35%	0%	10%
7 Reproducibility	70%	60%	40%	30%
8 298B Plan	75%	50%	40%	40%
References	90%	75%	70%	70%

The document measures how much of each written report section is finished. Sections 1–2 were submitted as Workbook 1 and are marked as fully documented. Sections 3.1–3.6 are documented at about 80–85% in docs/data_engineering.md on GitHub; the text still has to be moved into the Workbook 2 report. Sections 4.1–4.4 and 4.6 are documented partly, in Workbook 1, docs/metrics.md and docs/baseline_notes.md; their Workbook 2 text has not been written. Section 4.5 and Section 5 are not yet documented because Model A and the prototype are not built. Section 2.3 has a 50-point gap in the other direction (100% documented, 50% complete) because the resource plan is written, while compute access and environment setup remain unresolved. The Complete percentages for Sections 3.2–3.6 cover the Ext-CDM pipeline, including the patient-level split and leakage checks (split v2). They do not cover new data fields or model evaluation. The remaining percentages are team estimates. Workbook 2 Sections 3–4 are due on November 11. The data-access arrangement also changed after Report 1: each data user now holds individually approved access and their own copy, and sharing one member's download was dropped.

Part B Minutes of Meetings

Meeting 1: Team Meeting 1, professor track (October 8, 2026)

Time and date: Thursday, October 8, 2026, 4:40 PM - 5:00 PM
Location / media: Zoom Online
Participants: Ritika Mukesh Neema, Rajesh Paruchuri, Akashkumar Senthilkumar, Sneha Singh, Shruthi Thirukumaran

Section	Achievements	Issues	Recommended Solutions

1.1–1.5	Presented the goal, requirements and success criteria (validity ≥ 0.85, escape rate ≤ 0.05), the four-model deliverables, the literature survey and the Hager et al. baseline (54.9%).	The Hager result measures diagnostic accuracy and is not directly comparable to the project's plan-validity metric. Some literature references and evidence links still need consolidation.	Clearly distinguish the Hager diagnostic-accuracy baseline from the project's plan-validity results. Complete the literature references and link the supporting papers and repository evidence.
2.1–2.4	Presented the roles, tools, workflow, schedule and risks (Linear cycles, one assignee per issue, milestones to December 8).	Compute access, issue tracking, and responsibility updates still require follow-up. Some Linear issues need accurate labels, assignees, cycles, and status updates.	Update Linear after each meeting, maintain one primary assignee per issue, add workstream labels, and review the schedule at each cycle milestone.
3.1–3.6	Presented the cleaned 2,400-case dataset: 75 duplicate rows removed, 1,840 / 460 / 100 split, validation checks passing.	Credentialed MIMIC data cannot be stored in GitHub or shared between team members. Some rule-required clinical facts are not directly available in Ext-CDM and may require additional MIMIC-IV tables or notes.	Keep credentialed data outside GitHub, use individual authorized copies, use synthetic data for CI, and document which clinical facts can or cannot be derived from the available tables.
4.1–4.6	Presented Models A to D and the baseline plan.	Open items included GPU or HPC access for the 70B baseline, the local model for Model A, and the secondary-metric targets.	Confirm the compute environment, select and document the local model for Model A, finalize the secondary metrics, and record all baseline attempts and blockers.

Meeting 2: Team Meeting 1, ISA demonstration (October 8, 2026)

Time and date: Thursday, October 8, 2026, 4:00 PM - 4:20 PM

Location / media: Zoom Online

Participants: Ritika Mukesh Neema, Rajesh Paruchuri, Akashkumar Senthilkumar, Sneha Singh, Shruthi Thirukumaran

Section	Achievements	Issues	Recommended Solutions
3.1–3.6	Demonstrated the data pipeline: checksum verification, cleaning, validation checks passed on clean data and failed on a corrupted copy, and fixed splits were generated.	The pipeline demonstration used controlled data; credentialed data access and reproducibility across each member's environment still require continued verification. The pipeline evidence also needs to be transferred into Workbook 2.	Preserve checksum and corruption checks in the automated workflow, document the exact run commands and environment, and transfer the pipeline results and screenshots into Workbook 2 Sections 3.1–3.6.

Meeting 3: Internal team meeting (October 20, 2026)

Time and date: Tuesday, October 20, 2026, 6:00 PM

Location / media: In person (offline)

Participants: Ritika Mukesh Neema, Rajesh Paruchuri, Akashkumar Senthilkumar, Sneha Singh, Shruthi Thirukumaran

Linear board: https://linear.app/298a-team-6-neurosymbolic/team/T6/all

Section	Achievements	Issues	Recommended Solutions
1.2–1.3	Completed the v2.0 rule catalog, valid/invalid synthetic examples, Z3 encoding, and unit tests. (Linear: T6-15, T6-16, T6-18, T6-19)	Second-team-member review of the rule sources is still pending.	Sneha should obtain an independent review and record initials/date in docs/rule_signoff.mdbefore November 5.
2.1	Completed data-access documentation, provenance records, integrity checks, and restricted-data handling. (Linear: T6-5, T6-43 to T6-48)	Credentialed data cannot be stored in GitHub or shared between members.	Keep data outside the repository, use individual authorized copies, and use synthetic data for CI.
2.2–2.4	Completed the team schedule, Linear cycle tracking, reproducibility plan, and CI plan (Linear: T6-7, T6-42)	GPU access and the 70B baseline environment are unresolved.	Confirm GPU access and environment, and attempt the first baseline run, by October 28.
3.1–3.6	Completed collection, cleaning, transformation, validation, EDA, fixed splits, and patient-level leakage checks. (Linear: T6-49 to T6-52, T6-146, T6-147)	Workbook 2 Sections 3–4 still need to be written from the repository evidence.	Transfer the completed pipeline evidence into Workbook 2 before November 11.
4.1–4.6	Defined the verifier, metrics, rule catalog, and baseline protocol. (Linear: T6-20, T6-21, T6-23)	Model A and the Hager baseline have not yet been run.	Implement Model A and run the baseline during Cycle 5.
References	Added the repository, MIMIC documentation, Hager paper, rule catalog, and split documentation.	Linear issues have no labels yet, and a few finished issues still show Todo or In Review.	Add labels to each issue, move finished issues to Done, and link the GitHub commit on each.

Part C Work completed this period

Member	Issues closed	Labels worked	Key artifact link
Sneha Singh	(T6-15, T6-16, T6-18 to T6-21, T6-23, T6-47, T6-51, T6-56, T6-61, T6-90 to T6-95, T6-99 to T6-101)	model, eval	Git-https://github.com/Akashkumarsenthil/Clinical-Discharge-Treatment-Plan-Verification/blob/main/docs/rule_catalog.md
Ritika Mukesh Neema	(T6-45, T6-49, T6-59, T6-64, T6-147)	data	Git-https://github.com/Akashkumarsenthil/Clinical-Discharge-Treatment-Plan-Verification/blob/main/data/splits/ext_cdm_splits_v2.json
Rajesh Paruchuri	(T6-30, T6-44, T6-52, T6-57, T6-71, T6-146)	data, eval	Git-https://github.com/Akashkumarsenthil/Clinical-Discharge-Treatment-Plan-Verification/blob/main/docs/split_leakage_check.md
Akashkumar Senthilkumar	(T6-6, T6-43, T6-48, T6-63, T6-68, T6-69, T6-70)	data, infra	Git-https://github.com/Akashkumarsenthil/Clinical-Discharge-Treatment-Plan-Verification/blob/main/docs/data_collection.md

Shruthi Thirukumaran	(T6-42, T6-46, T6-50, T6-58, T6-66, T6-67)	infra, data	Git-https://github.com/Akashkumarsenthil/Clinical-Discharge-Treatment-Plan-Verification/tree/main/.github/workflows

Current code and results

The data pipeline (Ext-CDM v1.1, cleaned, validated and explored), the fixed patient-grouped split (v2: 1,840 / 460 / 100, seed 298, no patient in two splits), the rule catalog (v2.0, 12 rules), the Z3 verifier and the evaluation harness are implemented and tested (51 tests, all passing on October 21). Files: src/clinverify/; data/splits/ext_cdm_splits_v2.json; docs/rule_catalog.md; docs/split_leakage_check.md. Not started: Model A, the baseline reproduction, and the scheduled pipeline, warehouse and monitoring.

Part D Blockers, risks and next period

Blocker or risk	Impact	Owner	Next action and date
Rule catalog sign-off	Catalog v2.0 is complete and owner-signed. A second team member still has to read the rules against the sources, and two sources were read as abstracts only.	Sneha	Second reviewer signs off, then tag rules-v2.0-signed-off and freeze the rules before Nov 5, when the verifier work starts.
Compute and setup for the 70B baseline	The Hager baseline has not been run in the current project. Suitable GPU access and the required environment are still unresolved.	Ritika / Rajesh; environment: Shruthi	Confirm GPU/70B access and finalize the environment by October 28. Run the first baseline reproduction by October 28 and record the configuration, commit, runtime, and metrics.

Missing fields and split checks	The v2.0 rules need case facts (such as complicated disease, infection, biliary cause) and treatment concepts that are not stored as fields in Ext-CDM. MIMIC-IV tables (prescriptions, pharmacy, poe, emar, labevents, transfers) and the notes can supply them; see docs/data_dictionary.md. The patient-level overlap check is done: 5 patients crossed the first split and the current split has none.	Akash / Rajesh (proposed)	Map the rules' required case facts and treatment concepts (for example complicated vs uncomplicated, infection, biliary cause) from Ext-CDM and MIMIC-IV tables, and list any that cannot be derived. Target: Nov 5, before the first benchmark run.
Scheduled pipeline, warehouse and monitoring	The data pipeline, tests, fixed splits, notebook, verifier, rule catalog and CI are in the team GitHub repository. A scheduled run, a warehouse and monitoring are not built yet, and they are needed for the November 12 demo.	Shruthi (environment lead)	Save the cleaned tables in the chosen warehouse, schedule the pipeline run and add basic monitoring. Target: Nov 9

Next steps for team review:

1. **Sneha:** get a second team member to read and sign off rule catalog v2.0 against its cited sources, tag rules-v2.0-signed-off, and freeze the rules before the benchmark runs. Note the two sources that were read as abstracts only.
2. **Shruthi** and **Sneha,** with **Rajesh** handling evaluation: extend the verifier and harness tests (valid plans, rule violations, cases with missing information) and run the first benchmark on the labeled synthetic plans. The working target in Workbook 1 is November 5.
3. **Ritika** and **Akash:** implement Model A using the agreed input and plan format. Record the model, settings, and hardware used. The working target in Workbook 1 is November 5.
4. **Rajesh** and **Ritika,** with Shruthi supporting setup: confirm GPU access and the environment for the Hager baseline. Record the commands, attempts, and blockers, and agree on an intermediate deadline before the final report.
5. **Akash** and **Shruthi,** subject to team agreement: save the cleaned tables in the chosen warehouse and add scheduled runs and monitoring for the November 12 ISA demo. The team also needs to complete Workbook 2 Sections 3–4 by November 11.